Measure Theoretic Probability
Lesson 24
Expectation of Simple Random Variables: Properties, Limits and LOTUS

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Outline

1. Recap and Setting
2. More Basic Properties
3. Convergence of Means
4. Independence and $\mathbb{E}[XY]$
5. The Law of the Unconscious Statistician
6. Infinite Sums and Expectation
7. Summary and Exercises

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Where We Are

Simple random variables and expectation

On $(\Omega, \mathcal{F}, \mathbb{P})$, a simple random variable in **canonical form** is

$$X = \sum_{i=1}^{n} x_i \, I_{A_i}, \qquad x_i \text{ distinct}, \quad A_i = \{X = x_i\} \in \mathcal{F} \text{ partition } \Omega,$$

and its expectation is

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \, \mathbb{P}(A_i) = \sum_{i=1}^{n} x_i \, \mathbb{P}(X = x_i).$$

Proved earlier

- $\mathbb{E}[X]$ does not depend on the representation (partition, overlapping sets, extra 0-terms).
- **P1:** $X = 0$ a.s. $\Rightarrow \mathbb{E}[X] = 0$.
- **P2:** $X \geq 0$ a.s. $\Rightarrow \mathbb{E}[X] \geq 0$.
- **P3:** $X \geq 0$ a.s. and $\mathbb{E}[X] = 0 \Rightarrow X = 0$ a.s.

Plan for Today

1. Finish the list of basic properties (P4 to P8), including **linearity**.
2. **Convergence of means**: when does $X_n \xrightarrow{\mathbb{P}} X$ give $\mathbb{E}[X_n] \to \mathbb{E}[X]$? (Bounded case, and two counterexamples.)
3. **Independence** and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.
4. The **law of the unconscious statistician**: $\mathbb{E}[g(X)] = \sum g(x_i)\mathbb{P}(A_i)$.
5. **Infinite sums** inside expectations.

Why prove "obvious" facts?

All of these hold for general random variables too. We prove them now, from the measure-theoretic definition, because we will **reuse** them as building blocks. Expectation is not yet defined for non-simple variables, so "random variable" means **simple** random variable throughout this lesson.

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Property 4: $\mathbb{E}[X] > 0 \Rightarrow \mathbb{P}(X > 0) > 0$

Idea

If the average is positive, some value must be positive, and it must be taken with positive probability.

Proof

Write X in canonical form. If $\mathbb{E}[X] = \sum_i x_i \mathbb{P}(A_i) > 0$, at least one term is positive: $x_i \mathbb{P}(A_i) > 0$ for some i. Since $\mathbb{P}(A_i) \geq 0$, this forces

$$x_i > 0 \quad \text{and} \quad \mathbb{P}(A_i) > 0.$$

On A_i we have $X = x_i > 0$, so $A_i \subseteq \{X > 0\}$. By monotonicity,

$$\mathbb{P}(X > 0) \;\geq\; \mathbb{P}(A_i) \;>\; 0. \qquad \square$$

Why "$\geq$" and not "$=$"

$\{X > 0\}$ is the union of *all* the A_i with $x_i > 0$. We only needed one of them.

Adding Two Simple Random Variables

Setup

$X = \sum_{i=1}^{n} x_i I_{A_i}$ and $Y = \sum_{j=1}^{m} y_j I_{B_j}$, with $\{A_i\}$ and $\{B_j\}$ partitions of Ω in $\mathcal{F}$.

Where does ω go?

ω lies in exactly one A_i and exactly one B_j, i.e. in exactly one **intersection** $A_i \cap B_j$. There $X(\omega) = x_i$ and $Y(\omega) = y_j$. So

$$aX + bY = \sum_{i=1}^{n} \sum_{j=1}^{m} (a\,x_i + b\,y_j)\, I_{A_i \cap B_j}.$$

The sets $\{A_i \cap B_j\}$ form a partition of Ω in $\mathcal{F}$ (the **common refinement**).

Remember

To add simple random variables, **intersect their partitions**. The new representation need not be canonical (different (i, j) may give the same value), but by earlier Lessons, this does not matter.

Property 5: Linearity

Theorem

For simple X, Y and $a, b \in \mathbb{R}$: $\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$.

Proof

Use the common refinement from the previous frame and the definition of $\mathbb{E}$ (valid for any partition representation):

$$\begin{aligned}\mathbb{E}[aX + bY] &= \sum_{i,j} (ax_i + by_j)\,\mathbb{P}(A_i \cap B_j) \\ &= a \sum_i x_i \sum_j \mathbb{P}(A_i \cap B_j) + b \sum_j y_j \sum_i \mathbb{P}(A_i \cap B_j) \\ &= a \sum_i x_i\,\mathbb{P}(A_i) + b \sum_j y_j\,\mathbb{P}(B_j) = a\mathbb{E}[X] + b\mathbb{E}[Y].\end{aligned}$$

In the last line we used finite additivity: $A_i = \bigsqcup_j (A_i \cap B_j)$ and $B_j = \bigsqcup_i (A_i \cap B_j)$. □

By induction

$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i]$ for every **finite** sum. Infinite sums need extra care (last section).

Linearity: a Numerical Check

Example

$\Omega = [0,1]$ with Lebesgue measure. Let

$$X = 1 \cdot I_{[0,\frac{1}{2})} + 3 \cdot I_{[\frac{1}{2},1]}, \qquad Y = 2 \cdot I_{[0,\frac{1}{4})} + 4 \cdot I_{[\frac{1}{4},1]}.$$

Common refinement for $X + Y$

Set	X	Y	$X+Y$
$[0, \frac{1}{4})$	1	2	3
$[\frac{1}{4}, \frac{1}{2})$	1	4	5
$[\frac{1}{2}, 1]$	3	4	7

$$\mathbb{E}[X+Y] = 3 \cdot \tfrac{1}{4} + 5 \cdot \tfrac{1}{4} + 7 \cdot \tfrac{1}{2} = \tfrac{11}{2}.$$

Compare

$\mathbb{E}[X] = 1 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = 2$ and $\mathbb{E}[Y] = 2 \cdot \frac{1}{4} + 4 \cdot \frac{3}{4} = \frac{7}{2}$. Indeed $2 + \frac{7}{2} = \frac{11}{2}$. ✓

Properties 6, 7, 8

P6: $X = Y$ a.s. $\Rightarrow \mathbb{E}[X] = \mathbb{E}[Y]$

$X - Y = 0$ a.s., so $\mathbb{E}[X - Y] = 0$ by P1. By linearity $\mathbb{E}[X] - \mathbb{E}[Y] = 0$. □

P7 (monotonicity): $X \leq Y$ a.s. $\Rightarrow \mathbb{E}[X] \leq \mathbb{E}[Y]$

$Y - X \geq 0$ a.s., so $\mathbb{E}[Y - X] \geq 0$ by P2. By linearity $\mathbb{E}[Y] - \mathbb{E}[X] \geq 0$. □

P8: $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$

$|X|$ is simple (values $|x_i|$). Since $X \leq |X|$ and $-X \leq |X|$, P7 and linearity give

$$\mathbb{E}[X] \leq \mathbb{E}[|X|] \qquad \text{and} \qquad -\mathbb{E}[X] \leq \mathbb{E}[|X|].$$

Together: $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$. □

Remark

P8 is the simplest case of **Jensen's inequality**, which we meet later.

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The Question

Can we pass the limit through $\mathbb{E}$?

If $X_n \xrightarrow{\mathbb{P}} X$, is $\lim_n \mathbb{E}[X_n] = \mathbb{E}[X]$?

Convergence of means is a weak statement on its own

Many different sequences have the same mean as X without being close to X. We therefore need convergence in probability **plus** a control on how large the X_n can be.

Uniform boundedness

(X_n) is **uniformly bounded** if there is a constant M with

$$|X_n(\omega)| \leq M \quad \text{for all } n \text{ and all } \omega$$

(or for almost all ω, in the a.s. version).

Theorem (Bounded Convergence, Simple Case)

Theorem

Let X_n, X be simple random variables with $X_n \xrightarrow{\mathbb{P}} X$ and $|X_n| \le M$ a.s. for all n. Then

$$\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Strategy

Show the stronger statement $\mathbb{E}\big[|X_n - X|\big] \to 0$. This is enough, because by linearity and P8,

$$\big|\mathbb{E}[X_n] - \mathbb{E}[X]\big| = \big|\mathbb{E}[X_n - X]\big| \le \mathbb{E}\big[|X_n - X|\big].$$

Picture

Fix ε. The set A_n where X_n and X differ by at least ε has probability $\to 0$. On A_n the difference is at most $2M$, which is big, but the set is tiny. Off A_n the difference is below ε. Both pieces are small.

Step 0: The Limit Is Bounded Too

Claim

$|X| \leq M$ a.s.

Proof

Fix $\varepsilon > 0$. If $|X(\omega)| > M + \varepsilon$ and $|X_n(\omega)| \leq M$, then $|X_n(\omega) - X(\omega)| > \varepsilon$. So for every n,

$$\mathbb{P}\big(|X| > M + \varepsilon\big) \leq \mathbb{P}\big(|X_n - X| > \varepsilon\big) \xrightarrow[n\to\infty]{} 0.$$

Hence $\mathbb{P}(|X| > M + \varepsilon) = 0$ for every $\varepsilon > 0$. Taking $\varepsilon = 1/k$ and a countable union gives $\mathbb{P}(|X| > M) = 0$. □

Consequence

Almost surely, $|X_n - X| \leq |X_n| + |X| \leq 2M$ for all n (triangle inequality).

Proof of the Theorem

Step 1: split according to a set A_n

Fix $\varepsilon > 0$ and let $A_n = \{\omega : \ |X_n(\omega) - X(\omega)| \geq \varepsilon\}$. Since $X_n \xrightarrow{\mathbb{P}} X$, $\mathbb{P}(A_n) \to 0$.

Step 2: pointwise bound (a.s.)

Every ω is in A_n or in A_n^c, so

$$|X_n - X| = |X_n - X|\, I_{A_n} + |X_n - X|\, I_{A_n^c} \ \leq \ 2M\, I_{A_n} + \varepsilon.$$

(On A_n use the bound $2M$; on A_n^c the difference is $< \varepsilon$.)

Step 3: take expectations (P7 and linearity)

$$\mathbb{E}\big[|X_n - X|\big] \leq 2M\, \mathbb{E}[I_{A_n}] + \varepsilon = 2M\, \mathbb{P}(A_n) + \varepsilon.$$

Step 4: let $n \to \infty$, then $\varepsilon \to 0$

$\limsup_n \mathbb{E}[|X_n - X|] \leq 0 + \varepsilon$. Since $\varepsilon > 0$ was arbitrary and the left side is ≥ 0, $\mathbb{E}[|X_n - X|] \to 0$. By the strategy, $\mathbb{E}[X_n] \to \mathbb{E}[X]$. □

Counterexample 1: $\xrightarrow{\mathbb{P}}$ Without Boundedness

Setup

$\Omega = [0,1]$, Lebesgue measure, and $X_n = n^2\, I_{[0,1/n]}$ (add $0 \cdot I_{(1/n,1]}$ if you want a full partition).

It converges in probability to 0

For $\varepsilon > 0$ and n large enough that $n^2 > \varepsilon$,

$$\mathbb{P}(|X_n - 0| > \varepsilon) = \mathbb{P}\big([0, 1/n]\big) = \tfrac{1}{n} \to 0.$$

But the means do not converge

$$\mathbb{E}[X_n] = n^2 \cdot \tfrac{1}{n} = n \longrightarrow \infty, \qquad \text{while } \mathbb{E}[0] = 0.$$

Lesson

The sequence is **not** uniformly bounded ($\sup X_n = n^2$). A shrinking set can still carry a huge mass if the values are large enough. Boundedness cannot be dropped from the theorem.

Counterexample 2: Means Converge, but Not in Probability

Setup

Let $X_1, X_2, \ldots$ and X be independent, each equal to $+1$ or -1 with probability $\frac{1}{2}$. (Independence is defined later in this lesson.)

Means agree

$\mathbb{E}[X_n] = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0 = \mathbb{E}[X]$ for all n, so $\mathbb{E}[X_n] \to \mathbb{E}[X]$ trivially.

No convergence in probability

Take $\varepsilon = 2$. Since $|X_n - X| \in \{0, 2\}$, $|X_n - X| \geq 2$ iff the two differ:

$$\mathbb{P}(|X_n - X| \geq 2) = \mathbb{P}(X_n = 1, X = -1) + \mathbb{P}(X_n = -1, X = 1) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2} \not\to 0.$$

Conclusion

$X_n \xrightarrow{\mathbb{P}} X \not\Leftarrow \mathbb{E}[X_n] \to \mathbb{E}[X]$. Convergence of means is genuinely weaker.

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Independent Random Variables

Definition

X and Y are **independent** if the σ-fields $\sigma(X)$ and $\sigma(Y)$ are independent:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B) \qquad \text{for all } A \in \sigma(X),\ B \in \sigma(Y).$$

In words

Every event decided by X is independent of every event decided by Y.

Used below

If $X = \sum x_i I_{A_i}$ and $Y = \sum y_j I_{B_j}$ are in canonical form, then $A_i = X^{-1}(\{x_i\}) \in \sigma(X)$ and $B_j = Y^{-1}(\{y_j\}) \in \sigma(Y)$ (singletons are Borel). So independence of X, Y gives

$$\mathbb{P}(A_i \cap B_j) = \mathbb{P}(A_i)\,\mathbb{P}(B_j) \quad \text{for all } i, j.$$

Product of Independent Variables

Theorem

If simple X, Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$.

Proof

A product of indicators is an indicator of the intersection: $I_{A_i} I_{B_j} = I_{A_i \cap B_j}$. Hence

$$XY = \sum_{i,j} x_i y_j \, I_{A_i \cap B_j}.$$

The sets $A_i \cap B_j$ partition Ω (not necessarily canonical, but that is fine). So

$$\begin{aligned}\mathbb{E}[XY] &= \sum_{i,j} x_i y_j \, \mathbb{P}(A_i \cap B_j) = \sum_{i,j} x_i y_j \, \mathbb{P}(A_i)\mathbb{P}(B_j) \\ &= \Big(\sum_i x_i \mathbb{P}(A_i)\Big)\Big(\sum_j y_j \mathbb{P}(B_j)\Big) = \mathbb{E}[X]\,\mathbb{E}[Y]. \qquad \square\end{aligned}$$

Remark

The same proof shows $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$. Independence is used in exactly one place: the factorisation of $\mathbb{P}(A_i \cap B_j)$.

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The Idea, with an Example

Example

X takes $-1, 0, 1$ with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. Let $Y = X^2$.

Way 1: use the distribution of X (the "unconscious" way)

$$\mathbb{E}[X^2] = (-1)^2 \cdot \tfrac{1}{2} + 0^2 \cdot \tfrac{1}{4} + 1^2 \cdot \tfrac{1}{4} = \tfrac{3}{4}.$$

Way 2: find the distribution of Y first

$Y = 0$ with probability $\frac{1}{4}$ and $Y = 1$ with probability $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$, so

$$\mathbb{E}[Y] = 0 \cdot \tfrac{1}{4} + 1 \cdot \tfrac{3}{4} = \tfrac{3}{4}.$$

Law of the unconscious statistician (LOTUS)

Both ways agree, so we never need to find the distribution of $g(X)$: just apply g to the values of X.

Statement and Setup of the Proof

Theorem (LOTUS, simple case)

Let $X = \sum_{i=1}^{n} x_i I_{A_i}$ be in canonical form and $g : \mathbb{R} \to \mathbb{R}$ any function. Then $Y = g(X)$ is simple and

$$\mathbb{E}[g(X)] = \sum_{i=1}^{n} g(x_i)\,\mathbb{P}(A_i).$$

What Y looks like

On A_i, $Y = g(x_i)$. So $Y = \sum_i g(x_i) I_{A_i}$, but this need not be canonical: two different $x_i, x_{i'}$ may give $g(x_i) = g(x_{i'})$. Let $y_1, \ldots, y_m$ $(m \le n)$ be the **distinct** values of Y, and

$$B_j = \{Y = y_j\} = \bigcup_{i:\ g(x_i) = y_j} A_i \ \in \mathcal{F}.$$

This is a **disjoint** union, because the A_i are disjoint.

Proof of LOTUS

Compute $\mathbb{E}[Y]$ from the canonical form of Y

$$\mathbb{E}[Y] = \sum_{j=1}^{m} y_j \, \mathbb{P}(B_j) = \sum_{j=1}^{m} y_j \sum_{i:\, g(x_i)=y_j} \mathbb{P}(A_i) \quad \text{(finite additivity)}.$$

Replace y_j by $g(x_i)$ inside the inner sum

In the inner sum, $y_j = g(x_i)$, so

$$\mathbb{E}[Y] = \sum_{j=1}^{m} \sum_{i:\, g(x_i)=y_j} g(x_i) \, \mathbb{P}(A_i).$$

Merge the double sum

Each index $i \in \{1, \ldots, n\}$ belongs to **exactly one** group $\{i : g(x_i) = y_j\}$ (namely j with $y_j = g(x_i)$). So the double sum runs over every i exactly once:

$$\mathbb{E}[Y] = \sum_{i=1}^{n} g(x_i) \, \mathbb{P}(A_i). \qquad \square$$

LOTUS: the Proof on the Example

$X \in \{-1, 0, 1\}$ on A_1, A_2, A_3, $g(x) = x^2$

Values of Y: $y_1 = 0$, $y_2 = 1$ (so $m = 2$, $n = 3$). Groups of indices:

- $y_1 = 0$: only $i = 2$ (since $g(0) = 0$), so $B_1 = A_2$.
- $y_2 = 1$: $i = 1$ and $i = 3$ (since $g(-1) = g(1) = 1$), so $B_2 = A_1 \cup A_3$.

The two sums

$$\underbrace{0 \cdot \mathbb{P}(A_2) + 1 \cdot \big(\mathbb{P}(A_1) + \mathbb{P}(A_3)\big)}_{\sum_j y_j \mathbb{P}(B_j)} = \underbrace{g(-1)\mathbb{P}(A_1) + g(0)\mathbb{P}(A_2) + g(1)\mathbb{P}(A_3)}_{\sum_i g(x_i)\mathbb{P}(A_i)}.$$

Remark

Applying g can **merge** values (fewer terms, $m \leq n$) but never creates new ones. Merging is exactly what finite additivity handles.

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Why Infinite Sums Are Delicate

What we know

Finite sums pass through $\mathbb{E}$ (linearity plus induction). For **infinite** sums, $\mathbb{E}\left[\sum_{n=1}^{\infty} X_n\right] = \sum_{n=1}^{\infty} \mathbb{E}[X_n]$ can fail in general.

A sufficient (not necessary) condition

Suppose $X = \sum_{n=1}^{\infty} X_n$ almost surely (meaning the partial sums $Y_n = \sum_{i=1}^{n} X_i$ converge to X a.s.), where X and all X_n are simple. If the partial sums are uniformly bounded, i.e. $|Y_n| \leq M$ a.s. for all n, then

$$\mathbb{E}\left[\sum_{n=1}^{\infty} X_n\right] = \sum_{n=1}^{\infty} \mathbb{E}[X_n].$$

Scope

Here we assume X is simple so that $\mathbb{E}[X]$ is already defined. Once expectation is extended to general random variables, the same proof applies verbatim.

Proof

Step 1: reduce to a limit of partial sums

$Y_n = \sum_{i=1}^{n} X_i$ is a finite sum of simple random variables, so it is simple, and $Y_n \xrightarrow{\text{a.s.}} X$ by assumption.

Step 2: use the bounded convergence theorem

Almost sure convergence implies convergence in probability, so $Y_n \xrightarrow{\mathbb{P}} X$. Also $|Y_n| \leq M$ a.s. for all n. By the bounded convergence theorem,

$$\mathbb{E}[Y_n] \longrightarrow \mathbb{E}[X].$$

Step 3: finite linearity

$\mathbb{E}[Y_n] = \sum_{i=1}^{n} \mathbb{E}[X_i]$, hence

$$\mathbb{E}[X] = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{\infty} \mathbb{E}[X_i]. \qquad \square$$

Remark

The proof also shows that the series $\sum_i \mathbb{E}[X_i]$ converges. We did not assume it.

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Summary

Key results (all for simple random variables)

Result	Statement
P4	$\mathbb{E}[X] > 0 \Rightarrow \mathbb{P}(X > 0) > 0$
P5 (linearity)	$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$; add via the common refinement $\{A_i \cap B_j\}$
P6	$X = Y$ a.s. $\Rightarrow \mathbb{E}[X] = \mathbb{E}[Y]$
P7 (monotonicity)	$X \leq Y$ a.s. $\Rightarrow \mathbb{E}[X] \leq \mathbb{E}[Y]$
P8	$\lvert\mathbb{E}[X]\rvert \leq \mathbb{E}[\lvert X\rvert]$
Bounded convergence	$X_n \xrightarrow{\mathbb{P}} X$, $\lvert X_n\rvert \leq M \Rightarrow \mathbb{E}[X_n] \to \mathbb{E}[X]$
Counterexamples	$n^2 I_{[0,1/n]}$ (no bound); i.i.d. ± 1 (means converge, no $\xrightarrow{\mathbb{P}}$)
Independence	$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
LOTUS	$\mathbb{E}[g(X)] = \sum_i g(x_i)\mathbb{P}(A_i)$
Infinite sums	bounded partial sums $\Rightarrow \mathbb{E}[\sum X_n] = \sum \mathbb{E}[X_n]$

Exercises

1. Let X take values $-2, 0, 3$ with probabilities $\frac{1}{3}, \frac{1}{2}, \frac{1}{6}$. Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$ (using LOTUS) and $\mathbb{E}[|X|]$. Check P8.
2. Show that $\mathbb{E}[I_A] = \mathbb{P}(A)$ and use linearity to compute $\mathbb{E}[I_A + I_B]$ for $A, B \in \mathcal{F}$.
3. In the example on the linearity slide, compute $\mathbb{E}[XY]$ directly from the refinement table. Is $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$? Explain.
4. Show that P7 gives $\min_i x_i \le \mathbb{E}[X] \le \max_i x_i$.
5. For the sequence $X_n = I_{[0,1/n]}$ on $[0,1]$, show $X_n \xrightarrow{\mathbb{P}} 0$ and check the bounded convergence theorem.
6. For $X_n = n\, I_{[0,1/n]}$, compute $\mathbb{E}[X_n]$. Does $X_n \xrightarrow{\mathbb{P}} 0$? Is (X_n) uniformly bounded?
7. Let X and Y be independent, each ± 1 with probability $\frac{1}{2}$. Compute $\mathbb{E}[XY]$ and $\mathbb{E}[(X+Y)^2]$.
8. Let X take values $-1, 1$ with probability $\frac{1}{2}$ each and $g(x) = |x|$. Find the distribution of $g(X)$ and verify LOTUS.
9. In Step 0 of the bounded convergence proof, why do we need a *countable* union at the end?

Outlook

Next lesson

Approximating measurable functions by simple functions. We show that every nonnegative random variable is the limit of an increasing sequence of simple random variables.

Looking ahead

That approximation is how we will define $\mathbb{E}[X]$ for **general** (e.g. continuous) random variables, without ever needing probability density functions. The properties proved in Lessons 23 and 24 are exactly what makes the extension work.